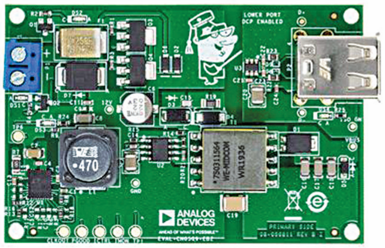
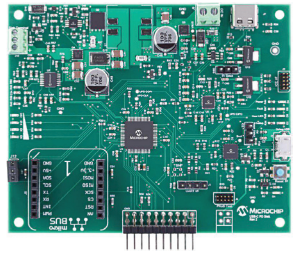
Interesting Reference Designs Of

Title of Reference Design	Dual USB Port Charger	USB Power Delivery Battery Charger
Image		
Key Attractions	The reference design for dual USB port charger can supply 2A at 5V from various direct current (DC) power sources. The design can handle reverse input up to 100V and operate safely up to 150°C.	The USB power delivery (PD) battery charger reference design includes a single-ended primary inductor converter (SEPIC) power supply, which complies with the 20V/5A 100W USB PD specification. The charger automatically shuts off when the charge current reaches a specified cutoff limit.
Highlights	The dual USB port charger CN0509 is a reference design by Analog Devices (ADI). The design supports a 2A, 5V output from different DC power sources and has a wide input voltage range. The circuit incorporates two USB charging ports, with one featuring a dedicated charging port (DCP) controller. The reference design can convert various DC power sources within the range of 5V to 100V into a regulated 5V, 2A power supply. Additionally, the design includes reverse voltage protection to safeguard against reverse voltage conditions. The design features two output ports that enable simultaneous use. The charger is based on an isolated flyback converter combined with a step-down (buck) DC-to-DC converter. The buck converter effectively converts higher DC voltages to lower voltages with minimal power loss and high power density. The design supports fast charging up to 2A and maintains consistent operation across input voltages ranging from 12V to 100V due to the constant 12V input of the isolated converter. The dual USB port charger is capable of handling reverse input connections of up to 100V. The design can operate safely at temperatures up to a maximum of 150°C.	The reference design of the USB power delivery battery charger by Microchip offers a simple solution to incorporate multi-cell Lithium-ion battery charging into existing applications. The reference design utilises the ATSAMD21J18A microcontroller, which is a 64-pin, low-power device equipped with a general-purpose ARM Cortex M0+ core. The design incorporates a Pro Kit on Board (PKoB) for USB programming and debugging purposes. The battery charger integrates a single-ended primary inductor converter (SEPIC) power supply, meeting the 20V/5A 100W USB PD specification. The charger employs a constant current/constant voltage charge algorithm, functioning through three main states: pre-conditioning, constant current charging, and constant voltage charging. The charger maintains constant current until near maximum battery voltage, then switches to constant voltage. The design periodically checks the battery voltage and adjusts the current, if necessary, to keep it at or slightly below the maximum voltage threshold. Once the charge current drops below a specified cutoff level, the charger will shut off but will remain active to monitor the battery.
Applications	The reference design is suitable for charging critical USB-powered devices such as mobile devices and portable gaming consoles.	The reference design can be used to develop high-quality USB power delivery chargers for smartphones, tablets, laptops, etc.
OEM Brand	Analog Devices (ADI)	Microchip
URL	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-a-dual-usb-port-charger	https://www.electronicsforu.com/electronics-projects/electronics-design-guides/reference-design-for-usb-power-delivery-battery-charger